



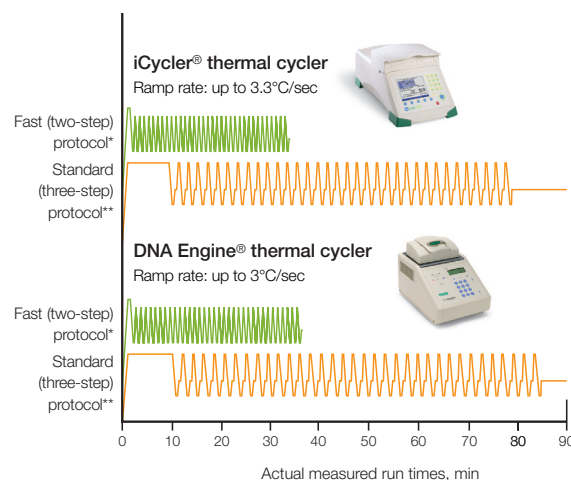
Thermal Cyclers and Reagents

Fast PCR With Bio-Rad Products

PCR run times can be dramatically reduced without giving up the flexibility, modularity, and gradient capability of Bio-Rad's thermal cyclers. With our cyclers, enzymes, and reaction vessels, you can:

- Run standard PCR reactions in only 35 minutes
- Obtain SYBR Green real-time PCR quantitation data in under 40 minutes
- Reduce run times for long (1–15 kb) targets 3- to 4-fold

For more information, visit us on the Web at
www.bio-rad.com/fastpcr/



* Fast (two-step) protocol, 98°C for 30 sec, then (92°C for 1 sec, 70°C for 20 sec) x 35 cycles.

** Standard (three-step) protocol, 95°C for 10 min, then (95°C for 15 sec, 60°C for 30 sec, 72°C for 30 sec) x 35 cycles, 72°C for 10 min.

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Easy Protocol Conversion Is the Key to Dramatically Reduced Run Times

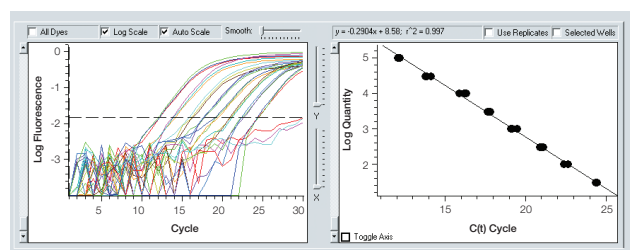
With standard Bio-Rad thermal cyclers, enzymes, and reaction vessels, a 1.5 hour PCR protocol can be reduced to 35 minutes:

- Shorten initial hot-start step with iTaq™ polymerase
- Combine annealing and extension steps at 70°C
- Reduce hold times for annealing/extension
- Ensure primer T_m values are close to 65°C

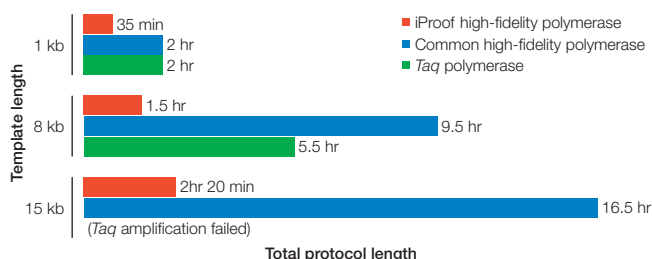
For details, download our fast PCR tutorial at www.bio-rad.com/fastpcr/



Fast two-step protocol results are comparable to those generated using standard protocols. β -Actin and β -globin targets were amplified from human genomic DNA using iTaq polymerase. Standard (S) protocol, 1.5 hr, fast (F) protocol, 35 min. For details see www.bio-rad.com/fastpcr/



SYBR Green qPCR in less than 40 min. Lambda phage target (150 bp) was amplified using iTaq™ SYBR® Green supermix. For details see www.bio-rad.com/fastpcr/



For long (1–15 kb) targets, use of iProof™ polymerase reduces run times 3- to 4-fold. Targets of 1, 8, or 15 kb were amplified using three different polymerases. A two-step PCR protocol was used with iProof polymerase; three-step protocols using the shortest recommended extension times were used with other polymerases. Because iProof polymerase requires an annealing temperature 5°C above typical annealing temperatures, two-step protocols often can be run without redesigning primers.

Ordering Information

Catalog #	Product Description
PTC-1148C	MJ Mini™ Gradient Thermal Cycler With Adjustable Heated Lid , holds 10 x 0.5 ml tubes, 48 x 0.2 ml tubes, or 48-well microplate; includes 120 x 0.2 ml 8-tube strips, 120 optical flat 8-cap strips, 1,000 x 0.2 ml individual tubes with attached flat caps
PTC-0200G	DNA Engine Thermal Cycler Chassis , does not include the Alpha™ unit (requires 1)
PTC-0220G	DNA Engine Dyad® Dual-Bay Thermal Cycler Chassis , does not include the Alpha units (requires 2)
PTC-0221G	Dyad Disciple™ Dual-Bay Thermal Cycler Chassis , does not include the Alpha units (requires 2)
PTC-0240G	DNA Engine Tetrad® 2 Thermal Cycler Chassis , does not include the Alpha units (requires at least 2 and fits up to 4)
170-9703	MyCycler™ Thermal Cycler System , includes MyCycler thermal cycler, PCR tubes, iTaq DNA polymerase, power cord, quick reference guide, instructions
170-8720	iCycler Thermal Cycler With 96-Well Reaction Module
170-8722	iCycler Thermal Cycler With 2 x 48 Dual Reaction Module
CFB-3120G	MiniOpticon™ Real-Time PCR Detector , includes optical housing, thermal cycler, and analysis software
CFB-3240	Chromo4™ Four-Color Real-Time PCR Detection System , includes optical housing, photonics shuttle, DNA Engine thermal cycler, 96-well sample block, analysis software
170-9770	MyiQ™ Single-Color Real-Time PCR Detection System , includes iCycler base, optics module, software CD-ROM, 96-well optical reaction module, optical-quality 96-well PCR plates, Microseal 'B' seals, communication cables, power cords, instructions
170-9780	iQ™5 Multicolor Real-Time PCR Detection System , includes iCycler base, optics module, software CD-ROM, 5 installed filter sets, 96-well reaction module, calibration solutions, optical-quality 96-well PCR plates, Microseal 'B' seals, communication cables, power cord, quick reference cards, instructions
170-8870	iTaq DNA Polymerase , 5 U/μl, includes 250 U polymerase, 1.25 ml 10x PCR buffer, 1.25 ml 50 mM MgCl ₂ solution
172-5301	iProof High-Fidelity DNA Polymerase , 2 U/μl, includes 100 U enzyme, 5x reaction buffers, MgCl ₂ solution, DMSO
172-5302	iProof High-Fidelity DNA Polymerase , 2 U/μl, includes 500 U enzyme, 5x reaction buffers, MgCl ₂ solution, DMSO
172-5310	iProof HF Master Mix , 100 x 50 μl reactions, includes 2x master mix, DMSO (for highest fidelity with most templates)
172-5311	iProof HF Master Mix , 500 x 50 μl reactions, includes 2x master mix, DMSO (for highest fidelity with most templates)
170-8880	iQ™ SYBR® Green Supermix , 100 x 50 μl reactions
170-8882	iQ SYBR Green Supermix , 500 x 50 μl reactions

Purchase of this instrument conveys a limited non-transferable immunity from suit for the purchaser's own internal research and development and for use in applied fields other than Human In Vitro Diagnostics under one or more of U.S. Patents Nos. 5,656,493, 5,333,675, 5,475,610 (claims 1, 44, 158, 160–163 and 167 only), and 6,703,236 (claims 1–7 only), or corresponding claims in their non-U.S. counterparts, owned by Applied Biosystems Corporation. No right is conveyed expressly, by implication or by estoppel under any other patent claim, such as claims to apparatus, reagents, kits, or methods such as 5' nuclease methods. Further information on purchasing licenses may be obtained by contacting the Director of Licensing, Applied Biosystems, 850 Lincoln Centre Drive, Foster City, California 94404, USA.

Bio-Rad's real-time thermal cyclers are licensed real-time thermal cyclers under Applied's United States Patent No. 6,814,934 B1 for use in research and for all other fields except the fields of human diagnostics and veterinary diagnostics.

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Appearances and specifications are subject to change without notice.

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